Comparative Analysis

Overview

This section compares the classical machine learning baseline (Part B) with the deep learning baselines (Part C) for automatic Kellgren–Lawrence (KL) knee osteoarthritis grading using X-ray images. Performance is evaluated using Accuracy, Macro F1-score, and Cohen's Kappa, along with qualitative analysis from confusion matrices.

Models Compared

Classical ML (Part B)

- Feature extraction: Gray-Level Co-occurrence Matrix (GLCM)
    - Contrast, Correlation, Energy, Homogeneity
    - Multiple distances and angles
- Classifier: Random Forest (500 trees)
- Class imbalance handling: `class_weight="balanced"`

Deep Learning (Part C)

1. **ResNet-18 (Transfer Learning)**
    - ImageNet pretrained
    - End-to-end CNN feature learning
2. **EfficientNet-B0 (Transfer Learning)**
    - ImageNet pretrained
    - Partial fine-tuning of higher feature blocks
    - WeightedRandomSampler to address class imbalance

Quantitative Performance Comparison

Model	Accuracy	Macro F1	Cohen Kappa
Classical ML (GLCM + RF)	**0.37**	**0.26**	–
ResNet-18	**0.63**	**0.64**	**0.49**
EfficientNet-B0	**0.60**	**0.60**	**0.45**

Which Approach Performed Better and Why?

Deep learning approaches significantly outperformed the classical ML baseline.

1. **Feature Learning Capability**
   - Classical ML relies on *hand-crafted texture features* (GLCM), which mainly capture local second-order statistics and cannot model complex anatomical patterns such as joint space narrowing, osteophytes, and bone deformation.
   - CNNs automatically learn hierarchical and spatially rich features, which are better suited for medical image interpretation.
2. **Transfer Learning Advantage**
   - Both ResNet-18 and EfficientNet-B0 leverage ImageNet pretraining, enabling robust low-level edge and texture representations even with limited medical data.
3. **Model Capacity vs Dataset Complexity**
   - Random Forests on fixed-length GLCM features underfit the problem.
   - CNNs adapt feature representations directly to KL grading.
4. **ResNet-18 vs EfficientNet-B0**
   - ResNet-18 is simpler and may generalize better on a moderate-sized dataset.
   - EfficientNet-B0, while more parameter-efficient, is more sensitive to hyperparameters and limited training epochs.

Confusion Matrix Analysis~ Hardest KL Grades to Predict

Across both classical ML and deep learning models, the most difficult classes were:

- **KL Grade 1 vs KL Grade 2**
- **KL Grade 2 vs KL Grade 3**

Adjacent KL grades differ by subtle radiographic changes. Even human experts show inter-observer variability for early OA grades. Thus Models frequently confuse mild OA with doubtful or moderate OA.

Observations:

- Extreme classes (**KL 0 and KL 4**) were predicted more reliably.
- Classical ML showed heavy bias toward majority mid-range classes.
- Deep learning models reduced confusion but did not eliminate it entirely.

Class Imbalance

The dataset exhibits a natural imbalance, with fewer samples for KL 0 and KL 4 compared to KL 2 and KL 3.

Impact on Classical ML:

- Despite `class_weight="balanced"`, the Random Forest still favored majority classes.
- Minority classes had very low recall, reducing Macro F1-score.

Impact on Deep Learning:

- Use of *WeightedRandomSampler* improved minority class exposure during training.
- Macro F1 and Cohen's Kappa indicate better balanced predictions.
- Remaining imbalance still affected borderline grades.

Conclusion

- Deep learning is clearly superior for KL grading from knee X-rays.
- ResNet-18 achieved the best overall performance, balancing accuracy and agreement.
- EfficientNet-B0 showed competitive results, but requires careful tuning and more data.
- Classical ML with GLCM features is insufficient for capturing the complexity of OA severity.